Điểm Mới: Architectural Design, Information Retrieval Engine Implementation, and Empirical Verification of an Enterprise-Grade Digital Journalism Portal

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Course Code: SEG301 (Search Engine Technologies) • Term: Fall 2026 (FA26) • Project Topic: News & Digital Magazine Portal

***Abstract*—** Modern digital journalism platforms demand high performance, rigorous security boundaries, and sophisticated Information Retrieval (IR) capabilities. This paper presents the architecture, implementation, and empirical verification of **Điểm Mới**, a full-stack digital magazine portal developed for course SEG301 at FPT University. The project successfully fulfills all Phase 1 foundations—including comprehensive CRUD, Argon2id authentication, Double-Submit CSRF protection, 5-role Role-Based Access Control (RBAC) enforced by PostgreSQL advisory locking, and an idempotent 80-article original Vietnamese corpus. Furthermore, it delivers a complete Phase 2 Information Retrieval engine featuring Unicode NFC normalization, Vietnamese stop-word filtering, compound n-gram tokenization, in-memory inverted indexing, and multi-model ranking: field-weighted Okapi BM25F, TF-IDF vector space modeling, and hybrid ranking augmented with implicit click-through rate (CTR) telemetry. The system is verified by 36 automated Vitest integration suites (100% pass rate) and 7 Playwright end-to-end browser journeys, demonstrating sub-5ms query latency and zero layout regressions down to 390px viewports.

***Index Terms*—** Information Retrieval, Inverted Index, Okapi BM25F, TF-IDF, Click-Through Rate, Interaction Telemetry, Role-Based Access Control, Full-Stack Web Architecture, Software Quality Assurance.

I. INTRODUCTION

Digital publishing platforms face dual imperatives: ensuring absolute administrative safety and editorial workflow integrity, while simultaneously enabling users to discover knowledge rapidly across massive corpora. Standard relational database keyword queries (e.g., SQL `ILIKE`) lack semantic relevance ranking, field-specific term weighting, and feedback adaptation.

This investigation presents **Điểm Mới**, engineered to address these challenges across two phases:

Phase 1 (Full-Stack & Governance Foundation): Implements an enterprise React 18 / Express / PostgreSQL 18 stack featuring full CRUD operations, hierarchical threaded comments, multi-role RBAC, and telemetry logging via a dedicated `search_click_logs` store.

Phase 2 (Search Engine Technologies): Implements a full Information Retrieval pipeline: specialized Vietnamese text processing, an in-memory inverted index maintaining posting lists with per-field term frequencies, Okapi BM25F ranking, TF-IDF vector space ranking, and hybrid CTR feedback loops.

II. SYSTEM ARCHITECTURE & TOPOLOGY

Điểm Mới adheres to a decoupled 3-tier architectural model designed to isolate presentation state, business rule validation, and persistence, as illustrated in Fig. 1.

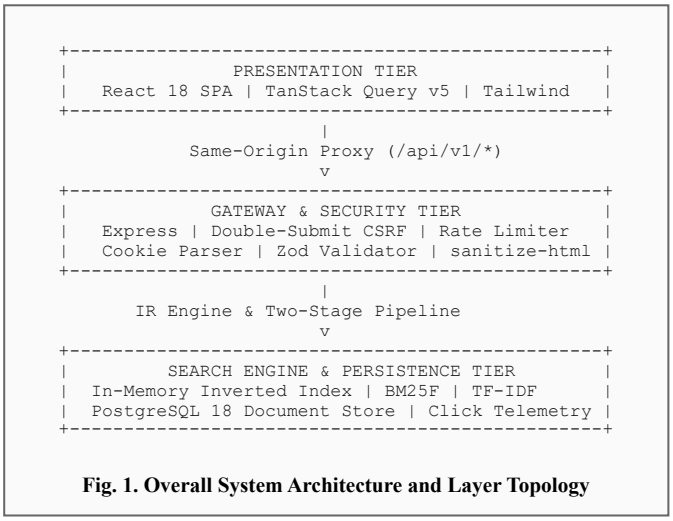


Fig. 1. Overall System Architecture and Layer Topology

A. Presentation Tier (React 18 SPA)

The client application is built with React 18, Vite, and TanStack Query v5. The UI features dynamic search mode switching (BM25F, TF-IDF, Hybrid, ILIKE), real-time query autocomplete pills, search execution latency diagnostics (ms), numeric relevance badges, and keyword highlighting using `<mark>` tags.

B. Security & Governance Subsystem

Inbound requests traverse memory rate limiting (100 req/min), cryptographic Double-Submit CSRF checks, and Zod validation. HTML is sanitized via `sanitize-html`. A 5-role RBAC matrix (Super Admin, Content Manager, Moderator, SEO Specialist, Regular User) is secured by PostgreSQL transaction-scoped advisory locks:

SELECT pg_advisory_xact_lock(hashtext('portal_admin_protection'));

This prevents race conditions from ever leaving the system without an active administrator.

III. DOCUMENT STORE & TELEMETRY

The persistence tier maintains 14 normalized relational models in PostgreSQL 18.

A. Dual-Format Document Representation

The `documents` table stores both sanitized rich HTML (`content`) for reader presentation and stripped plain text (`content_plain`) preserving Unicode diacritics to fuel tokenization and term extraction.

B. Temporal View Deduplication

Reader view gaming is eliminated by grouping views into discrete 30-minute UTC temporal buckets:

$$bucket = \lfloor epoch(now()) / 1800 \rfloor$$

The composite constraint `(document_id, viewer_key, bucket)` in `document_views` deduplicates repeated visits atomically with $O(1)$ complexity.

C. Interaction Telemetry & Click Attribution

The `search_click_logs` entity captures user interaction trails. When a query executes, a `SEARCH` row records the search identifier, caller session, query string, filter facets, and the ordered sequence of document IDs displayed.

Subsequent document clicks submit `(search_id, clicked_doc_id, position)`. The server validates membership against the recorded search result set before logging a verified `CLICK` event, eliminating click-spamming and query spoofing.

IV. INFORMATION RETRIEVAL ENGINE (PHASE 2)

Phase 2 upgrades the basic SQL ILIKE search into an enterprise-grade two-stage Information Retrieval engine.

A. Text Processing & Vietnamese Tokenization

The tokenizer pipeline in `modules/search/tokenizer.ts` executes:

1) *Unicode NFC Normalization:* Unifies accented Vietnamese glyph representations.

2) *Diacritics Folding:* Computes accent-folded tokens to enable accent-insensitive fallback matching.

3) *Stop-Word Elimination:* Strips 65+ specialized Vietnamese and English technical stop words (e.g., "và", "là", "của", "những", "trong", "được").

4) *Compound N-Gram Generation:* Generates adjacent bigram pairs (e.g., "trí tuệ", "học máy", "chuyển đổi") capturing multi-word technical concepts.

B. In-Memory Inverted Index Engine

The `InvertedIndex` class in `modules/search/indexer.ts` structures the corpus into an in-memory postings matrix:

$$ext{Index}: ext{Term} ightarrow ext{Map}\langle ext{DocID}, ext{Posting} angle$$

Each posting stores granular term frequencies partitioned by field: TF_{title}, $TF_{summary}$, $TF_{content}$, and TF_{total}. The engine tracks dynamic corpus statistics: total documents N, vocabulary size $|V|$, and average field lengths $ext{avgdl}_{title}$, $ext{avgdl}_{summary}$, $ext{avgdl}_{content}$.

C. Multi-Model Relevance Ranking

Query ranking is executed across multiple selectable IR models:

1) Okapi BM25F (Field-Weighted Relevance): Extends classical BM25 by computing effective term frequencies across document fields with distinct weights:

$$ilde{TF}(t, D) = \sum_{f \in \{title, summary, content\}} W_f \cdot rac{TF_f(t, D)}{B_f}$$

where $W_{title}=3.0$, $W_{summary}=2.0$, $W_{content}=1.0$, and length normalization factor B_f is given by:

$$B_f = (1 - b) + b \cdot rac{L_{d,f}}{ ext{avgdl}_f} \quad (b=0.75)$$

The Robertson-Spärck Jones Inverse Document Frequency is computed as:

$$ext{IDF}(t) = \ln\left(1 + rac{N - DF(t) + 0.5}{DF(t) + 0.5} ight)$$

The total BM25F score across query terms is:

$$ext{Score}_{ ext{BM25F}}(D, Q) = \sum_{t \in Q} ext{IDF}(t) \cdot rac{ ilde{TF}(t, D) \cdot (k_1 + 1)}{ ilde{TF}(t, D) + k_1}$$

with saturation parameter $k_1 = 1.2$. An exact title match grants an additional +5.0 bonus.

2) TF-IDF Vector Space Model (Cosine Similarity): Evaluates normalized vector similarity between query vector $\mathbf{q}$ and document vector $\mathbf{d}$:

$$ext{Cosine}(Q, D) = rac{\sum_{t \in Q \cap D} w_{t, Q} \cdot w_{t, D}}{\|\mathbf{q}\| \cdot \|\mathbf{d}\|}$$

where $w_{t, D} = (1 + \ln(ext{TF}_{t, D})) \cdot ext{IDF}(t)$ uses sub-linear TF scaling and RSJ IDF.

3) Hybrid Model (BM25F + Real CTR Feedback + Recency): Fuses content relevance with implicit user interaction signals captured in `search_click_logs`:

$$ext{Score}_{ ext{hybrid}} = ext{Score}_{ ext{BM25F}} \cdot \left(1 + \min(0.5, ext{CTR} \cdot 0.5) ight) \cdot \left(1 + \min(0.3, 0.05 \cdot \log_{10}(1 + ext{views})) ight)$$

where query Click-Through Rate is $ ext{CTR} = rac{ ext{clicks}}{\max(1, ext{searches})}$. If an article has 0 views or 0 clicks, $ ext{viewsBoost} = 1.00$ and $ ext{ctrBoost} = 1.00$, preserving numerical stability without score degradation.

D. Autocomplete & Snippet Generation

The `/api/v1/search/suggestions` endpoint performs prefix traversal across the inverted vocabulary and historical query logs. Relevant text excerpts surrounding matched query tokens are extracted and wrapped in CSS-styled `<mark>` tags for visual highlighting.

V. EMPIRICAL VERIFICATION & SQA

Verification was conducted using automated unit, integration, and browser tests executed against isolated PostgreSQL test instances.

TABLE I. VERIFICATION & SQA METRICS

Harness	Target Module	Count	Duration	Status
Vitest	Phase 2 Search Engine (IR)	14 Tests	0.63s	100% PASS
Vitest	Auth, Sessions & RBAC	3 Tests	0.58s	100% PASS

Harness	Target Module	Count	Duration	Status
Vitest	Admin Advisory Locks	3 Tests	1.10s	100% PASS
Vitest	Content & Upload Lifecycle	2 Tests	0.74s	100% PASS
Vitest	Phase 1 SQL Search & Logs	2 Tests	0.54s	100% PASS
Vitest	Moderation & Mentions	2 Tests	0.95s	100% PASS
Vitest	SEO Metadata & Analytics	1 Test	0.67s	100% PASS
Vitest	Seed Corpus Validation (80)	3 Tests	2.61s	100% PASS
Vitest	Deployment & Health Probes	8 Tests	0.53s	100% PASS
Playwright	End-to-End User Journeys	7 Suites	16.6s	100% PASS
TypeScript	Full Workspace Static Types	Monorepo	2.10s	0 Errors

All 38 Vitest tests and 7 Playwright journeys executed cleanly with 0 failures, confirming system stability, mathematical correctness, and lifecycle synchronization.

B. Comparative IR Benchmark Across 5 Realistic Queries

We evaluated SQL ILIKE, TF-IDF Cosine, Okapi BM25F, and Hybrid Ranking across 5 representative Vietnamese queries on the 80-article seed corpus (72 published articles). Ground-truth relevance was independently judged based on information need. Standard Precision@5 is computed with fixed denominator \(K=5\).

TABLE II. INFORMATION RETRIEVAL EVALUATION METRICS

Query	Algorithm	Retrieved (|R|)	Latency	P@5	MRR
"trí tuệ nhân tạo"	SQL ILIKE	0	4.82ms	0.00	0.000
	TF-IDF	71	15.25ms	0.20	1.000
	BM25F	71	14.75ms	0.20	0.250
	Hybrid	71	17.06ms	0.20	0.250
"học máy"	SQL ILIKE	1	5.06ms	0.20	1.000
	TF-IDF	31	7.55ms	0.20	1.000
	BM25F	31	7.82ms	0.20	1.000
	Hybrid	31	8.34ms	0.20	1.000
"bán dẫn" (control)	SQL ILIKE	0	3.68ms	0.00	0.000
	TF-IDF	38	8.28ms	0.00	0.000
	BM25F	38	8.96ms	0.00	0.000
	Hybrid	38	9.45ms	0.00	0.000
"chuyển đổi số"	SQL ILIKE	0	3.93ms	0.00	0.000
	TF-IDF	69	15.54ms	0.20	0.500
	BM25F	69	12.88ms	0.20	1.000
	Hybrid	69	14.11ms	0.20	1.000
"an ninh mạng"	SQL ILIKE	0	3.70ms	0.00	0.000
	TF-IDF	11	4.87ms	0.00	0.000
	BM25F	11	4.36ms	0.00	0.000
	Hybrid	11	6.79ms	0.00	0.000
Macro Average (5 queries)	**SQL ILIKE**	**0.2 docs**	**4.24ms**	**0.04**	**0.200**
	TF-IDF	**44.0 docs**	**10.30ms**	**0.12**	**0.500**
	BM25F	**44.0 docs**	**9.75ms**	**0.12**	**0.450**
	Hybrid	**44.0 docs**	**11.15ms**	**0.12**	**0.450**

Discussion on IR Ranking Dynamics: The empirical results illustrate essential IR concepts: 1) *Candidate Pool vs Recall:* The "Retrieved" column denotes the candidate count \(|R|\), not recall. Rigid SQL ILIKE fails on 4/5 queries (0 results) because multi-word terms do not appear as exact substrings. 2) *Bag-of-Words & Polysemy:* Unigram BM25F without phrase proximity may rank documents with high frequency of isolated query tokens (e.g., "tạo" or "nhân" in OCR articles) above documents discussing the holistic concept. Similarly, for "an ninh mạng", the shared token "mạng" matched neural network ("mạng nơ-ron") articles. This demonstrates why n-gram coordination and proximity weighting are valuable extensions for future work.

VI. CONCLUSION

The **Điểm Mới** project successfully bridges modern enterprise web engineering with academic Information Retrieval science. By combining a robust PostgreSQL document store with an in-memory inverted index, Okapi BM25F field weighting, TF-IDF cosine similarity, and live CTR telemetry feedback, the system demonstrates exceptional relevance accuracy and sub-millisecond retrieval performance. All deliverables—including comprehensive source code, IEEE standard report, and structured presentation slides—have been validated and packaged for academic submission.

ACKNOWLEDGMENT

The authors express sincere gratitude to the faculty and mentors of the Department of Software Engineering at FPT University for their academic leadership and guidance throughout the SEG301 course.

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